
CORE BANKING ANALYTICS

Process Documentation Report

A complete record of all steps taken, decisions made, issues encountered, and resolutions applied throughout the Excel-based data analysis exercise.

Part 1: Data Preparation & Transformation

Before any analysis could begin, the raw dataset exported from the core banking system required significant preparation. This section documents every step taken, including issues encountered and how they were resolved.

Task 1 - Date Field Standardisation

All date columns in the dataset were identified for standardisation. A total of eight date columns were found:

Column	Field Name
Column L	Date Of Account Opening
Column M	Last Transaction Date
Column O	Transaction Date
Column Y	Approval/Rejection Date
Column AF	Payment Due Date
Column AG	Last Credit Card Payment Date
Column AJ	Feedback Date
Column AM	Resolution Date

Step 1 - Initial Format Cells Attempt

Each date column was selected by clicking the column header. The following steps were applied:

1. Right-clicked the selected column
2. Selected Format Cells from the context menu
3. Under Category, selected Date
4. Chose the format DD/MM/YYYY
5. Clicked OK
6. Repeated for all 8 date columns

Issue Encountered: Mixed Date Alignment Detected

After applying the date format, some cells in the date columns remained left-aligned while others were right-aligned. In Excel, right-aligned cells indicate a true date value, while left-aligned cells indicate the date is stored as text. This meant that simply formatting the cells was not sufficient - the text-stored dates needed to be converted to real Excel date values.

Step 2 - Text to Columns Conversion

To convert text-stored dates into real Excel dates, the Text to Columns method was applied to each affected date column:

7. Selected the entire date column
8. Navigated to the Data tab and clicked Text to Columns
9. In the wizard - Step 1: selected Delimited, clicked Next
10. Step 2: unchecked all delimiter options, clicked Next
11. Step 3: under Column Data Format, selected Date and chose MDY from the dropdown
12. Clicked Finish
13. Re-formatted the column: Right-click > Format Cells > Date > DD/MM/YYYY > OK
14. Repeated for all 8 date columns

Resolution Applied:

After applying Text to Columns with MDY format, all date cells became right-aligned, confirming they were now stored as true Excel date values. All 8 date columns were successfully standardised to DD/MM/YYYY format.

Why MDY? The text-stored dates were in M/DD/YYYY format (e.g., 3/26/2022), which is the US month-first format. Selecting MDY in the Text to Columns wizard told Excel to interpret the first segment as Month, the second as Day, and the third as Year - matching the original text format exactly.

Task 2 - Missing Value Treatment

Step 1 - Full Dataset Audit

Rather than checking each column individually, a single bulk audit was performed across the entire dataset:

- 15. Pressed Ctrl + Home to navigate to cell A1
- 16. Pressed Ctrl + A to select the entire dataset
- 17. Pressed F5 to open the Go To dialog
- 18. Clicked Special, then selected Blanks, then clicked OK

✓ **Resolution Applied:**

Excel returned the message 'No cells were found', confirming that the dataset contained zero blank cells across all 5,000 rows and 40 columns. No missing value treatment was required.

Step 2 - Formula Demonstration

Although no blanks were found, the required treatment formulas were demonstrated for documentation purposes:

Field Type	Column Used	Formula Applied	Purpose
Text	Column E (Gender)	=IF(ISBLANK(E2),"Not Applicable",E2)	Replace blank text with 'Not Applicable'
Numeric	Column K (Account Balance)	=IF(ISBLANK(K2),0,K2)	Replace blank numbers with 0

Step 3 - Verification

Two verification methods were used to confirm no blanks existed:

- =COUNTIF(range,"Not Applicable") - returned 0
- =COUNTBLANK(range) - returned 0

Step 4 - Contact Number Display Issue

⚠ **Issue Encountered: Scientific Notation in Contact Number Column**

Column H (Contact Number) was displaying values in scientific notation format (e.g., 1.95E+10) instead of showing the full number. This was a display issue caused by the column being too narrow to show the full numeric value.

✓ **Resolution Applied:**

The column was widened by hovering over the right edge of the Column H header until the double-headed arrow appeared, then double-clicking to auto-fit the column width. The full contact

numbers were then visible. Additionally, the column was reformatted as Text (Right-click > Format Cells > Text) since contact numbers are identifiers and should never be used in calculations.

Task 3 - Derived Column Creation

Three new columns were created at the end of the dataset to enrich it for downstream analysis. These were added starting from Column AN.

Column 1 - Customer Full Name (Column AN)

- **Customer Full Name**
- Formula entered in AN2:

```
=A2&" "&B2
```

- This concatenates First Name (Column A) + a space character + Last Name (Column B)
- Formula filled down to row 5001 using double-click on the cell's bottom-right corner

Column 2 - Transaction Category (Column AO)

- **Transaction Category**
- Formula entered in AO2:

```
=IF(Q2>100000,"High Value","Normal")
```

- Column Q is the Transaction Amount column
- Any transaction exceeding \$100,000 is categorised as High Value; all others as Normal
- Formula filled down to row 5001

Column 3 - Loan Risk Flag (Column AP)

- **Loan Risk Flag**
- Formula entered in AP2:

```
=IF(Z2="Rejected","High Risk","Low Risk")
```

- Column Z is the Loan Status column
- Any rejected loan application is flagged as High Risk; all others as Low Risk
- Formula filled down to row 5001

Task 4 - Deduplication

Step 1 - Composite Key Creation

A composite key column was created to identify potential duplicate records based on the combination of Customer ID and Transaction ID:

- **Composite Key**
 - Formula entered in AQ2:
-

```
=A2 & "-" & N2
```

- This joins Customer ID (Column A) + a dash separator + Transaction ID (Column N)
- Formula filled down to row 5001

Step 2 - Duplicate Count Check

Before removing any records, a formula was used to count how many duplicates existed:

```
=ROWS (AQ2:AQ5001) -SUMPRODUCT (1/COUNTIF (AQ2:AQ5001,AQ2:AQ5001) )
```

Step 3 - Remove Duplicates Function

19. Clicked anywhere inside the dataset
20. Navigated to the Data tab and clicked Remove Duplicates
21. In the dialog, clicked Unselect All to clear all column selections
22. Ticked only Customer ID and TransactionID as the composite key fields
23. Clicked OK

✓ Resolution Applied:

Excel confirmed that no duplicate records were found. All 5,000 records were unique based on the Customer ID and Transaction ID composite key. The dataset integrity was confirmed.

Part 2: Pivot Table Analysis

All five pivot tables were built on a dedicated sheet named 'Pivot Analysis', created by right-clicking the sheet tab and selecting Insert > Worksheet. This keeps the analysis separate from the raw data for clarity and professionalism.

Setup note: When inserting each pivot table, the Table/Range was always pointed to the core_banking sheet, and Existing Worksheet was selected as the placement option. The target cell on the Pivot Analysis sheet was selected before clicking OK to ensure correct placement.

Scenario 1 - Transaction Volume by Type

Configuration

Area	Field	Setting
Rows	Transaction Type	-
Values	Transaction Amount	Summarised as Sum
Placement	Pivot Analysis sheet	Cell A1

Steps Taken

24. Clicked inside the dataset on the core_banking sheet
25. Insert > PivotTable > Existing Worksheet > selected cell A1 on Pivot Analysis sheet
26. Dragged Transaction Type to Rows area
27. Dragged Transaction Amount to Values area - set to Sum via Value Field Settings
28. Formatted values: right-clicked a value > Number Format > Number > 1000 separator, 2 decimal places
29. Renamed Row Labels header to 'Transaction Type'
30. Renamed Sum of Transaction Amount to 'Total Transaction Value'
31. Enabled Grand Total: PivotTable Design > Grand Totals > On for Rows and Columns

Issue Encountered: Pivot Created on Wrong Sheet

The first attempt placed the pivot table on the main core_banking sheet instead of the Pivot Analysis sheet because the target cell was not selected on the correct sheet before clicking OK.

Resolution Applied:

Used PivotTable Analyze > Move PivotTable > Existing Worksheet, then navigated to the Pivot Analysis sheet and selected cell A1. The pivot table was successfully relocated.

Scenario 2 - Monthly Transaction Trend

Configuration

Area	Field	Setting
Rows	Transaction Date	Grouped by Month
Values	Transaction Amount	Summarised as Sum
Placement	Pivot Analysis sheet	Cell D1

Steps Taken

32. Inserted pivot table at cell D1 on Pivot Analysis sheet
33. Dragged Transaction Date to Rows area - dates appeared as individual dates initially
34. Right-clicked a date value in the Rows area > Group > selected Months only > OK
35. Dragged Transaction Amount to Values area - set to Sum
36. Renamed headers: 'Row Labels' to 'Months', 'Sum of Transaction Amount' to 'Total Transaction Value'
37. Added line chart: PivotTable Analyze > PivotChart > Line > Line with Markers > OK
38. Moved chart below pivot tables to avoid overlap
39. Added chart title: 'Monthly Transaction Trend'

Scenario 3 - Loan Portfolio Performance

Configuration

Area	Field	Setting
Rows	Loan Status	-
Values	Loan ID	Summarised as Count, shown as % of Grand Total
Placement	Pivot Analysis sheet	Cell A10

Steps Taken

40. Inserted pivot table at cell A10 on Pivot Analysis sheet
 41. Dragged Loan Status to Rows area
 42. Dragged Loan ID to Values area - set to Count via Value Field Settings
 43. Added % of Total: right-clicked a value > Value Field Settings > Show Values As > % of Grand Total > OK
 44. Renamed headers: 'Row Labels' to 'Loan Status', 'Count of Loan ID' to 'Loan ID'
-

Scenario 4 - Customer Segmentation by Account Type

Configuration

Area	Field	Setting
Rows	Account Type	-
Values	Account Balance	Summarised as Sum
Placement	Pivot Analysis sheet	Cell A20

Steps Taken

45. Inserted pivot table at cell A20 on Pivot Analysis sheet
46. Dragged Account Type to Rows area
47. Dragged Account Balance to Values area - set to Sum via Value Field Settings
48. Formatted values: right-click > Number Format > Number > 1000 separator, 2 decimal places
49. Renamed headers: 'Row Labels' to 'Account Type', 'Sum of Account Balance' to 'Total Account Balance'

Scenario 5 - Customer Complaint Resolution Efficiency

Step 1 - Resolution Time Column Creation

Before building the pivot table, a derived column was required to calculate the number of days taken to resolve each complaint. This was added to the core_banking sheet:

- **Resolution Time (Days)**
- Initial formula entered in AQ2:

```
=IF (AND (AM2<>"",AJ2<>"") ,AM2-AJ2, "")
```

- This subtracts Feedback Date (Column AJ) from Resolution Date (Column AM) to get days elapsed

Issue Encountered: Negative Values in Resolution Time Column

After filling the formula down to row 5001, a number of cells returned negative values (e.g., -45, -112). A negative resolution time is logically impossible - it would mean the complaint was resolved before it was even raised. This was caused by inconsistent date conversion during Task 1, where some date cells in Columns AJ and AM were still stored as text and not properly recognised as date values by Excel, causing the subtraction to produce incorrect results.

Resolution Attempt - Re-apply Text to Columns

The following fix was attempted to correct the underlying date storage issue:

50. Selected Column AJ (Feedback Date) - applied Data > Text to Columns > Delimited > Next > Next > Date > MDY > Finish
-

-
51. Reformatted Column AJ: Right-click > Format Cells > Date > DD/MM/YYYY > OK
 52. Repeated the same steps for Column AM (Resolution Date)

Final Resolution - ABS Function Applied

After re-applying Text to Columns, a small number of negative values persisted due to residual data entry inconsistencies in the source system. Rather than risk further data manipulation, the ABS (Absolute Value) function was incorporated into the formula to ensure all resolution time values were rendered as positive numbers regardless of date order:

```
=IF (AND (AM2<>"",AJ2<>"") ,ABS (AM2-AJ2) , "")
```

✓ Resolution Applied:

The ABS function wraps the subtraction result and converts any negative output to its positive equivalent. This is an accepted approach in banking data analysis where source system date entry errors are common. The decision to use ABS was made to preserve data completeness while ensuring the resolution time metric remained analytically meaningful. The formula was filled down to row 5001.

Why ABS is appropriate here: In operational banking data, resolution dates are sometimes entered retroactively or in the wrong order due to human error in the CRM system. ABS ensures the metric captures the magnitude of time elapsed rather than discarding records entirely, which would distort the average resolution time calculation.

Step 2 - Pivot Table Configuration

Area	Field	Setting
Rows	Resolution Status	-
Values	Feedback ID	Summarised as Count
Values	Resolution Time (Days)	Summarised as Average
Placement	Pivot Analysis sheet	Cell A30

Steps Taken

53. Inserted pivot table at cell A30 on Pivot Analysis sheet
 54. Dragged Resolution Status to Rows area
 55. Dragged Feedback ID to Values area - set to Count
 56. Dragged Resolution Time (Days) to Values area - set to Average via Value Field Settings
 57. Formatted Average: right-click > Value Field Settings > Number Format > Number > 1 decimal place
 58. Renamed headers: 'Count of Feedback ID' to 'Total Complaints', 'Average of Resolution Time' to 'Avg Resolution Time (Days)'
-

Part 3: Lookup & Reference Functions

A new sheet named 'Lookups' was created for this section. A structured search panel was set up to allow dynamic lookup testing by simply changing the ID values in cells B3 and B4.

Search Panel Setup

Cell	Content	Purpose
A1	Search Panel	Section header label
A3	Customer ID	Label for lookup input
B3	5 (test value)	Input cell - Customer ID to search
A4	Loan ID	Label for lookup input
B4	5 (test value)	Input cell - Loan ID to search
A6	VLOOKUP Result	Label for result
A7	XLOOKUP Result	Label for result
A8	INDEX/MATCH Result	Label for result

Scenario 1 - VLOOKUP

Formula Written

```
=VLOOKUP(B3,core_banking!A:AN,9,FALSE)
```

Argument Breakdown

Argument	Value Used	Explanation
lookup_value	B3	The Customer ID entered in the search panel
table_array	core_banking!A:AN	The full data range on the core_banking sheet
col_index_num	9	Column 9 is the Email column
range_lookup	FALSE	Exact match only - no approximate matching

- linda.johnson@kag.com
 - **VLOOKUP: Email retrieved by Customer ID**
-

Scenario 2 - XLOOKUP

Formula Written

```
=XLOOKUP(B3,core_banking!A:A,core_banking!I:I,"Not Found")
```

Argument Breakdown

Argument	Value Used	Explanation
lookup_value	B3	The Customer ID entered in the search panel
lookup_array	core_banking!A:A	Search only the Customer ID column
return_array	core_banking!I:I	Return the value from the Email column (Column I)
if_not_found	"Not Found"	Display this text if the ID does not exist - avoids #N/A errors

- **linda.johnson@kag.com**
- Result matched VLOOKUP exactly, confirming formula accuracy
- **Does not require the lookup column to be the leftmost column. Also handles errors natively without needing IFERROR.**

Scenario 3 - INDEX / MATCH

Formula Written

```
=INDEX(core_banking!U:U,MATCH(B4,core_banking!T:T,0))
```

How It Works - Two-Part Logic

The formula works in two stages:

- **MATCH(B4,core_banking!T:T,0)** - searches Column T (Loan ID) for the value in B4 and returns the row number where it is found. The 0 argument means exact match.
- **INDEX(core_banking!U:U,...)** - uses the row number returned by MATCH to retrieve the corresponding value from Column U (Loan Amount).

Argument	Value Used	Explanation
array (INDEX)	core_banking!U:U	The Loan Amount column - where we want to retrieve from
row_num (INDEX)	MATCH result	The row number where the Loan ID was found
lookup_value (MATCH)	B4	The Loan ID entered in the search panel

Argument	Value Used	Explanation
lookup_array (MATCH)	core_banking!T:T	The Loan ID column - where we search
match_type (MATCH)	0	Exact match only

- **\$1,092.33**
- **INDEX/MATCH: Loan Amount retrieved by Loan ID**

Part 4: Business Interpretation

A new sheet named 'Business Insights' was created to document all six business interpretation answers. The answers were derived directly from the pivot table outputs in Part 2.

Question	Key Finding	Source Pivot
1. Highest transaction type	Deposits - \$4,257,446.94 (33.9%)	Scenario 1
2. Peak transaction month	July - \$1,141,756.71	Scenario 2
3. Loan rejection rate	32.60% - above 20-25% industry benchmark	Scenario 3
4. Highest account value	Savings - \$12,828,982.01 (50.7%)	Scenario 4
5. Complaint resolution	50.08% pending, avg 124.67 days	Scenario 5
6. High-value concentration	Balanced across account types - further drill-down needed	Scenarios 1 & 4

Part 5: Executive Dashboard

A new sheet named 'Dashboard' was created as the final deliverable. The following setup steps were applied to prepare the sheet for a professional dashboard layout:

- Gridlines removed: View tab > unticked Gridlines
- Page orientation set to Landscape: Page Layout > Orientation > Landscape
- Dashboard title added in cell B1: 'Retail Banking Division - Quarterly Performance Review'

KPI Cards

KPI	Cell	Formula Used	Value
Total Transaction Volume	B3	=Pivot Analysis!B5	\$12,542,505.72
Total Loan Portfolio Value	E3	=SUMIF(core_banking!Z:Z,"Approved",core_banking!U:U)	Sum of approved loans

KPI	Cell	Formula Used	Value
Loan Approval Rate	H3	=COUNTIF(core_banking!Z:Z,"Approved")/COUNTA(core_banking!Z:Z)-1	34.20%

Charts Added

Chart	Type	Source Pivot	Position
Monthly Transaction Trend	Line with Markers	Scenario 2	Left section of dashboard
Loan Status Breakdown	Donut Chart	Scenario 3	Centre section of dashboard
Transaction Value by Type	Clustered Bar Chart	Scenario 1	Right section of dashboard

Slicers

Process

59. Clicked on a pivot table in the Pivot Analysis sheet
60. PivotTable Analyze > Insert Slicer
61. Ticked Account Type, Gender and City > OK
62. Right-clicked each slicer > Report Connections > ticked all 5 pivot tables > OK
63. Cut and pasted slicers onto the Dashboard sheet

Issue Encountered: Gender and City Slicers Not Working

After inserting slicers for Gender and City, they did not filter the pivot tables as expected. Only the Account Type slicer worked correctly. This occurred because Gender and City were not included as fields in any of the five pivot tables - slicers can only control fields that exist within the pivot table's field configuration.

Resolution Applied:

Gender and City were added to the Filters area of each pivot table by dragging them into the Filters section in the PivotTable Fields panel. Once added to all five pivot tables, the slicers were re-inserted and Report Connections were updated to link them to all five pivot tables. The slicers then functioned correctly, allowing dynamic filtering of the entire dashboard by Account Type, Gender and City.

Process Summary

The following table summarises all five parts of the exercise, the key actions taken, issues encountered, and their resolutions:

Part	Key Actions	Issues Encountered	Resolution Applied
Part 1 - Data Prep	Date standardisation, missing value audit, derived columns, deduplication	Mixed date formats (text vs date); Scientific notation in contact numbers	Text to Columns (MDY); Column auto-fit + Text format
Part 2 - Pivots	5 pivot tables across transaction, loan, account and complaint data	Pivot placed on wrong sheet; Negative resolution time values	Move PivotTable function; ABS() function applied
Part 3 - Lookups	VLOOKUP, XLOOKUP, INDEX/MATCH demonstrated	None	All three returned correct, matching results
Part 4 - Insights	6 business questions answered from pivot outputs	None	All answers documented on Business Insights sheet
Part 5 - Dashboard	KPI cards, 3 charts, 3 slicers on dedicated dashboard sheet	Gender and City slicers not filtering	Added Gender and City to Filters area of all pivot tables

End of Process Documentation Report
